DEBUGGING W/ FLOXGLOVE

Friday, July 05, 2024 11:37 AM

1. Prerequisites:
   a. Camera Data: We can publish debug images from "b3rb_ros_line_follower/b3rb_ros_line_follower/b3rb_ros_edge_vectors.py".
      i. Create a new publisher in the __init__ function as shown in the attached image. We need to create a new topic and in our case we have chosen "/b3rb_ros_line_follower/vector_image".

```python
class EdgeVectorsPublisher(Node):
        def __init__(self):
                super().__init__('edge_vectors_publisher')

                self.subscription_camera = self.create_subscription(
                        CompressedImage,
                        '/camera/image_raw/compressed',
                        self.camera_image_callback,
                        10)

                self.publisher_edge_vectors = self.create_publisher(
                        EdgeVectors,
                        '/edge_vectors',
                        10)

                self.publisher_thresh_image = self.create_publisher(
                        CompressedImage,
                        "/b3rb_ros_line_follower/thresh_image",
                        0)

                self.publisher_vector_image = self.create_publisher(
                        CompressedImage,
                        "/b3rb_ros_line_follower/vector_image",
                        0)

```

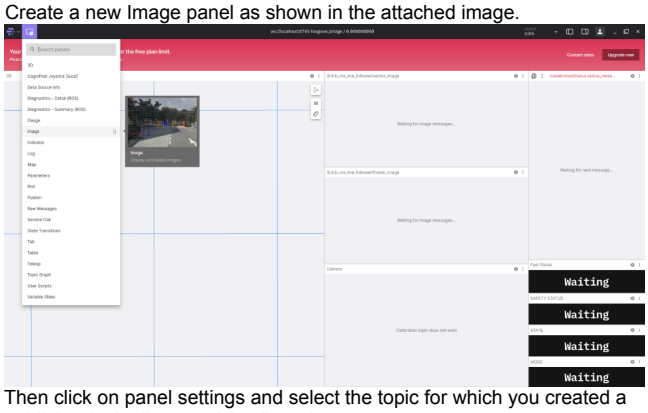
      ii. Now, pass this publisher (self.publisher_vector_image) and the image as arguments to the "**publish_debug_image**" function whenever you want to output your debug images.
      iii. cd ~/cognipilot/cranium
      iv. colcon build
      v. source install/setup.bash
      vi. Add the same topic name "/b3rb_ros_line_follower/vector_image" to the default_value of topic_whitelist in electrode/src/electrode/launch/electrode.launch.py as shown in the attached image.

```python
    DeclareLaunchArgument('topic_whitelist',
        default_value=['["/b3rb_ros_line_follower/vector_image","/camera/image_raw/compressed","/camera/camera_info","/cerebri/out/status","/cerebri/out/nav_sat_fix","/global_costmap/costmap","/map","global_costmap/published_footprint","/plan","/robot_description","/tf"]'],
        description='topic_whitelist for foxglove'
    ),
```

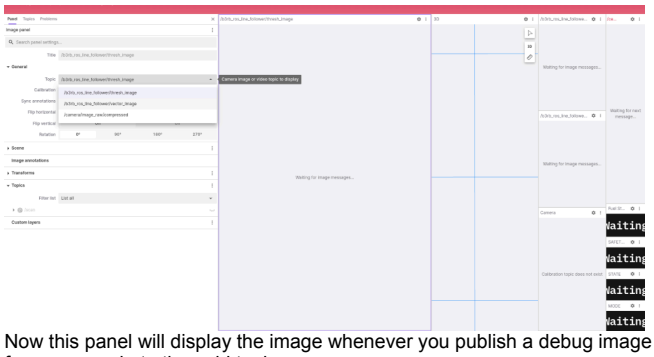
      vii. cd ~/cognipilot/electrode
      viii. colcon build
      ix. source install/setup.bash
   b. LIDAR Data:
      i. Add "/scan" to the default_value of topic_whitelist in electrode/src/electrode/launch/electrode.launch.py as shown in the attached image.

```python
    DeclareLaunchArgument('topic_whitelist',
        default_value=['["/camera/image_raw/compressed","/camera/camera_info","/cerebri/out/status","/cerebri/out/nav_sat_fix","/global_costmap/costmap","/map","global_costmap/published_footprint","/plan","/robot_description","/tf","/b3rb_ros_line_follower/thresh_image","/b3rb_ros_line_follower/vector_image","/scan"]'],
        description='topic_whitelist for foxglove'
    ),
```

      ii. cd ~/cognipilot/electrode
      iii. colcon build
      iv. source install/setup.bash
2. Launching Foxglove:
   a. ros2 launch electrode electrode.launch.py sim:=True
   b. Camera Data:
      i. Create a new Image panel as shown in the attached image.



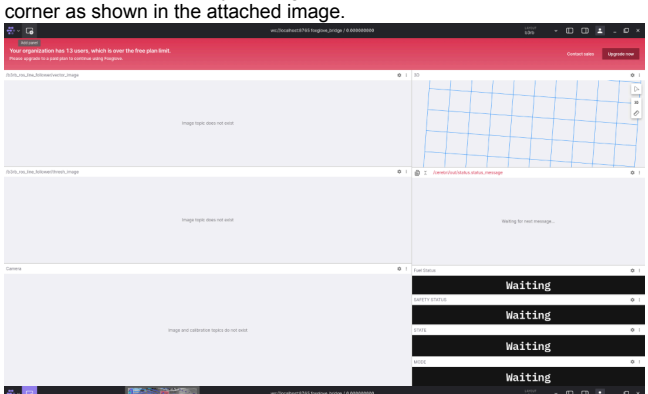
      ii. Then click on panel settings and select the topic for which you created a publisher in the "prerequisites" section.

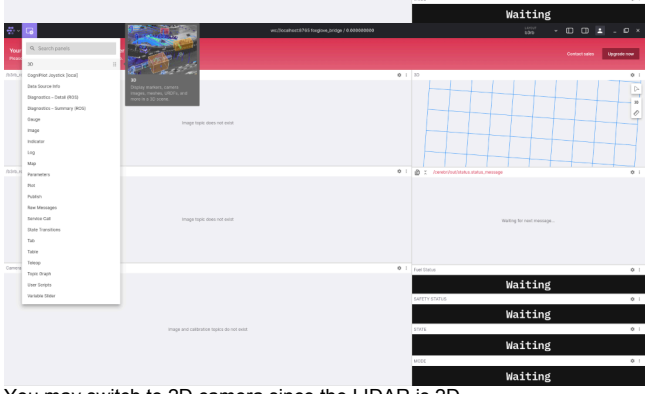


iii. Now this panel will display the image whenever you publish a debug image from your code to the said topic.

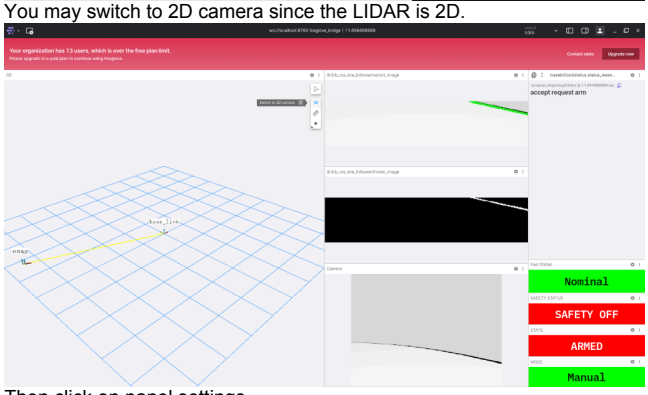
c. LIDAR Data:

i. Create a new 3D panel by clicking on the add panel button from the top-left corner as shown in the attached image.

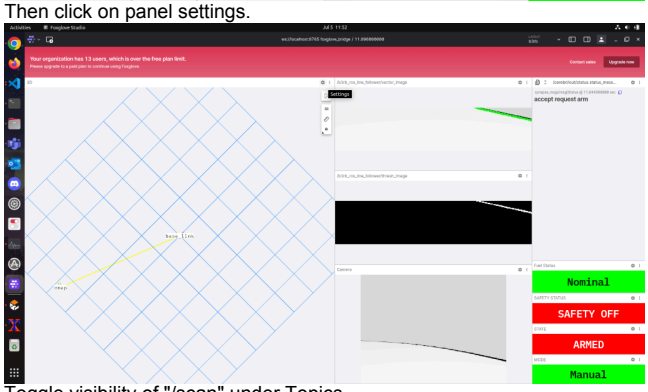




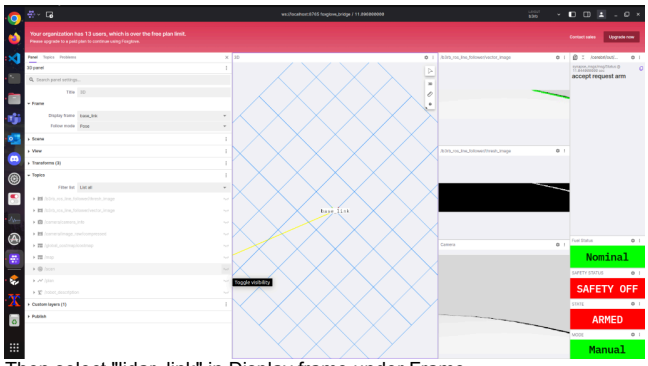
ii. You may switch to 2D camera since the LIDAR is 2D.



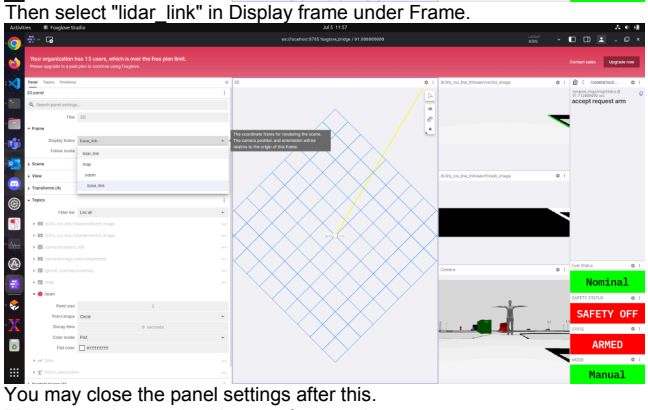
iii. Then click on panel settings.



iv. Toggle visibility of "/scan" under Topics.

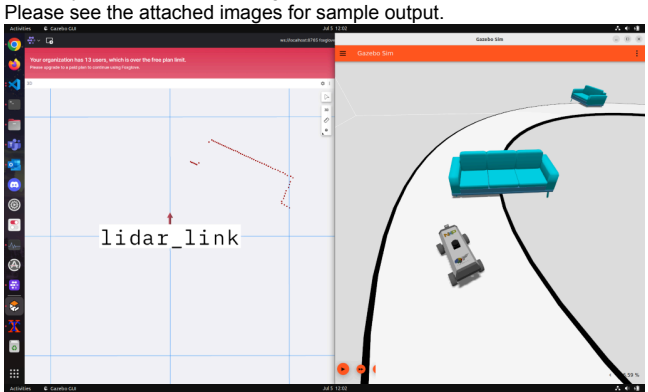


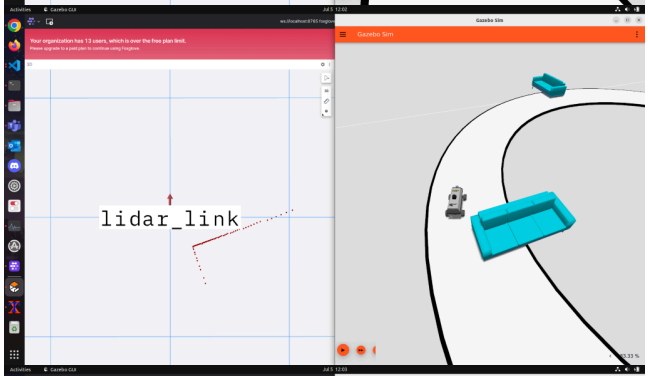
v. Then select "lidar_link" in Display frame under Frame.

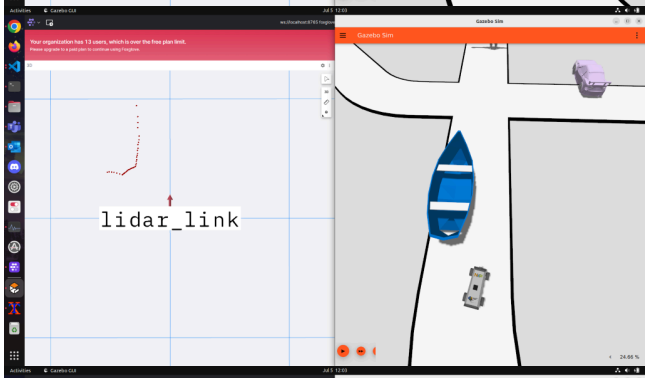


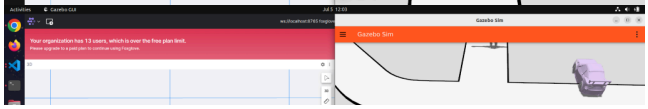
vi. You may close the panel settings after this.

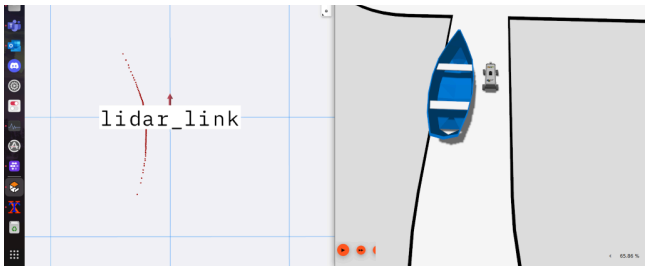
vii. Please see the attached images for sample output.










lidar_link